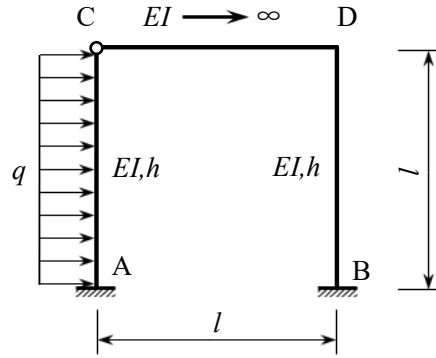




解(1)基本未知量

由于 CD 杆 EI 无穷大故 D 处无转角

所以基本未知量只有 C 处的水平位移 Δ_1

(2)杆端弯矩

固端弯矩查表求得

荷载作用下

$$M_{AC} = \frac{qh^2}{8}$$

位移作用下

$$M_{AC} = \frac{3i}{h}$$

$$M_{BD} = M_{DB} = \frac{6i}{h}$$

D 点弯矩平衡

$$M_{DC} = -\frac{6i}{h}$$

水平方向力平衡

$$k_{11} = \frac{3i}{h^2} + \frac{12i}{h^2} = \frac{15i}{h^2}$$

$$k_{11} * \Delta_1 + F_{1p} = 0$$

$$\Delta_1 = \frac{qh^3}{30i}$$

对左侧柱

$$M_{\overline{F}} = \frac{qh^2}{8} + \frac{3i}{h} * \frac{qh^3}{30i} = \frac{9qh^2}{40}$$

对右侧柱

$$M_{\perp} = M_{\overline{F}} = \frac{6i}{h} * \frac{qh^3}{30i} = \frac{qh^2}{5}$$

$$M_{DC} = \frac{qh^2}{5}$$

最终弯矩图如下:

